Fourier $X(\omega) = \mathcal{F}x(t) = \int x(t)e^{-j\omega t}dt$ $X(f) = \int x(t)e^{-j2\pi f t}dt$

Inverse Fourier $x(t) = \mathcal{F}^{-1}X(\omega) = \frac{1}{2\pi}\int x(t)e^{j\omega t}dt$ $x(t) = \int X(f)e^{j2\pi f t}dt$

$X^{(Hz)}$ 即 $X^{(rad)}$ 把 ω 换成 $2\pi f$

另 $\delta(\omega)$ 要换成 $\frac{1}{2\pi}\delta(f)$

Prop. $x(t-t_0) \longleftrightarrow X(\omega)e^{-j\omega t_0}$

$x(t)e^{j\omega_0 t} \longleftrightarrow X(\omega-\omega_0)$

$X(t) \longleftrightarrow 2\pi x(\omega)$ $X(t) \longleftrightarrow x(f)$

$x_1 * x_2 \longleftrightarrow X_1 \cdot X_2$

Fourier

$x(t)$	$X(\omega)$	$X(f)$
1	$2\pi\delta(\omega)$	$\delta(f)$
$\delta(t)$	1	1
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	$\delta(f-f_0)$
$\delta(t-t_0)$	$e^{j\omega t_0}$	$e^{-j2\pi f t_0}$
$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	$\frac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$
$\sin\omega_0 t$	$\frac{\pi}{j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$	$\frac{1}{2j}[\delta(f-f_0)-\delta(f+f_0)]$
$e^{-\lambda t}u(t)$	$\frac{1}{j\omega+\lambda}$	$\frac{1}{j2\pi f+\lambda}$
$te^{-\lambda t}u(t)$	$(\frac{1}{j\omega+\lambda})^2$	
$e^{-\lambda\lvert t\rvert}$	$\frac{2\lambda}{\omega^2+\lambda^2}$	
$\frac{2\lambda}{t^2+\lambda^2}$	$2\pi e^{-\lambda\lvert\omega\rvert}$	$2\pi e^{-\lambda 2\pi\lvert f\rvert}$
Rect	Sinc	
Triangle	Sinc^2	

时频性质

Time Domain	Freq Domain	
Discrete	Periodic	CFS : Cnt Prd - Dct Apd
Continous	Aperiodic	CFT : Cnt Apd - Cnt Apd
Periodic	Discrete	DTFT : Dct Apd - Cnt Prd
Aperiodic	Continous	DFT : Dct Prd - Dct Prd